

Q1. Write a program to check whether a given number is present in an array or not (Linear search).

```
#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter number of elements: ";
    cin >> n;

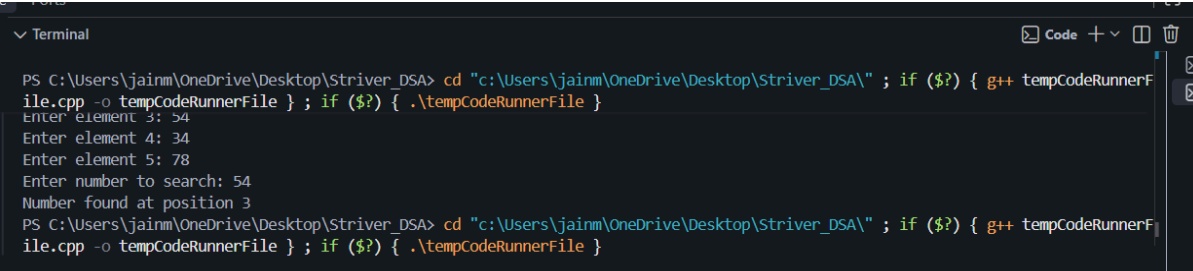
    int arr[100];
    for (int i = 0; i < n; i++) {
        cout << "Enter element " << i + 1 << ": ";
        cin >> arr[i];
    }

    int key;
    cout << "Enter number to search: ";
    cin >> key;

    int found = 0;
    for (int i = 0; i < n; i++) {
        if (arr[i] == key) {
            found = 1;
            cout << "Number found at position " << i + 1 << endl;
            break;
        }
    }

    if (found == 0) {
        cout << "Number not found in array" << endl;
    }

    return 0;
}
```



```
PS C:\Users\jainm\OneDrive\Desktop\Striver_DSA> cd "c:\Users\jainm\OneDrive\Desktop\Striver_DSA\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter element 3: 54
Enter element 4: 34
Enter element 5: 78
Enter number to search: 54
Number found at position 3
PS C:\Users\jainm\OneDrive\Desktop\Striver_DSA> cd "c:\Users\jainm\OneDrive\Desktop\Striver_DSA\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
```

Q2: Write a program to get second maximum and second minimum elements in an array.

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int n;
```

```
    cout << "Enter number of elements: ";
```

```
    cin >> n;
```

```
    int arr[100];
```

```
    for (int i = 0; i < n; i++) {
```

```
        cout << "Enter element " << i + 1 << ": ";
```

```
        cin >> arr[i];
```

```
    }
```

```
    int max1 = -1000000, max2 = -1000000;
```

```
    int min1 = 1000000, min2 = 1000000;
```

```
    for (int i = 0; i < n; i++) {
```

```
        if (arr[i] > max1) {
```

```
            max2 = max1;
```

```
            max1 = arr[i];
```

```
        } else if (arr[i] > max2 && arr[i] != max1) {
```

```
            max2 = arr[i];
```

```
        }
```

```
        if (arr[i] < min1) {
```

```
            min2 = min1;
```

```
            min1 = arr[i];
```

```
        } else if (arr[i] < min2 && arr[i] != min1) {
```

```
            min2 = arr[i];
```

```
        }
```

```
    }
```

```
    cout << "Second maximum element is: " << max2 << endl;
```

```
    cout << "Second minimum element is: " << min2 << endl;
```

```

    return 0;
}

```



```

Terminal
PS C:\Users\jainm\OneDrive\Desktop\Striver_DSA> cd "c:\Users\jainm\OneDrive\Desktop\Striver_DSA" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter element 1: 34
Enter element 2: 23
Enter element 3: 56
Enter element 4: 67
Second maximum element is: 56
Second minimum element is: 34
PS C:\Users\jainm\OneDrive\Desktop\Striver_DSA>

```

Q3. Write a program to perform insertion (any location), deletion (any location) and traversal in an array.

```

#include <iostream>
using namespace std;

```

```

int main() {
    int arr[100];
    int n;
    cout << "Enter number of elements: ";
    cin >> n;

    for (int i = 0; i < n; i++) {
        cout << "Enter element " << i + 1 << ": ";
        cin >> arr[i];
    }

    int choice;
    do {
        cout << "\n1. Insert\n2. Delete\n3. Traverse\n4. Exit\n";
        cout << "Enter your choice: ";
        cin >> choice;

        if (choice == 1) {
            int pos, value;
            cout << "Enter position to insert (1 to " << n + 1 << "): ";
            cin >> pos;
            cout << "Enter value to insert: ";
            cin >> value;

            for (int i = n; i >= pos; i--) {
                arr[i] = arr[i - 1];
            }
            arr[pos - 1] = value;

```

```

        n++;
        cout << "Element inserted successfully" << endl;
    }
    else if (choice == 2) {
        int pos;
        cout << "Enter position to delete (1 to " << n << "): ";
        cin >> pos;

        for (int i = pos - 1; i < n - 1; i++) {
            arr[i] = arr[i + 1];
        }
        n--;
        cout << "Element deleted successfully" << endl;
    }
    else if (choice == 3) {
        cout << "Array elements: ";
        for (int i = 0; i < n; i++) {
            cout << arr[i] << " ";
        }
        cout << endl;
    }
    else if (choice == 4) {
        cout << "Exiting program" << endl;
    }
    else {
        cout << "Invalid choice" << endl;
    }
} while (choice != 4);

return 0;
}

```

```
Terminal
PS C:\Users\jainm\OneDrive\Desktop\Striver_DSA> cd "c:\Users\jainm\OneDrive\Desktop\Striver_DSA\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter number of elements: 4
Enter element 1: 45
Enter element 2: 23
Enter element 3: 78
Enter element 4: 87

Terminal
PS C:\Users\jainm\OneDrive\Desktop\Striver_DSA> cd "c:\Users\jainm\OneDrive\Desktop\Striver_DSA\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter element 4: 87

1. Insert
2. Delete
3. Traverse
4. Exit
Enter your choice: 2

Terminal
PS C:\Users\jainm\OneDrive\Desktop\Striver_DSA> cd "c:\Users\jainm\OneDrive\Desktop\Striver_DSA\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
2. Delete
3. Traverse
4. Exit
Enter your choice: 2
Enter position to delete (1 to 4): 3
Element deleted successfully
```

Q4. Write a menu driven program to perform addition, multiplication and subtraction of 2 arrays.

```
#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter size of arrays: ";
    cin >> n;

    int a[100], b[100], result[100];
```

```

cout << "Enter elements of first array:" << endl;
for (int i = 0; i < n; i++) {
    cin >> a[i];
}

cout << "Enter elements of second array:" << endl;
for (int i = 0; i < n; i++) {
    cin >> b[i];
}

int choice;
cout << "\n1. Addition\n2. Multiplication\n3. Subtraction\n";
cout << "Enter your choice: ";
cin >> choice;

if (choice == 1) {
    for (int i = 0; i < n; i++) {
        result[i] = a[i] + b[i];
    }
    cout << "Result of addition: ";
}
else if (choice == 2) {
    for (int i = 0; i < n; i++) {
        result[i] = a[i] * b[i];
    }
    cout << "Result of multiplication: ";
}
else if (choice == 3) {
    for (int i = 0; i < n; i++) {
        result[i] = a[i] - b[i];
    }
    cout << "Result of subtraction: ";
}
else {
    cout << "Invalid choice" << endl;
    return 0;
}

for (int i = 0; i < n; i++) {
    cout << result[i] << " ";
}
cout << endl;

return 0;

```

}

```
Terminal
PS C:\Users\jainm\OneDrive\Desktop\Striver_DSA> cd "c:\Users\jainm\OneDrive\Desktop\Striver_DSA\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter size of arrays: 4
Enter elements of first array:
23
54
67
78
Enter elements of second array:
```

Q5. Write a program to perform sorting while merging (Merge two sorted arrays into one sorted array).

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
    int n1, n2;
    cout << "Enter size of first sorted array: ";
    cin >> n1;

    int a[100];
    cout << "Enter elements in sorted order:" << endl;
    for (int i = 0; i < n1; i++) {
        cin >> a[i];
    }

    cout << "Enter size of second sorted array: ";
    cin >> n2;

    int b[100];
    cout << "Enter elements in sorted order:" << endl;
    for (int i = 0; i < n2; i++) {
        cin >> b[i];
    }

    int merged[200];
    int i = 0, j = 0, k = 0;

    while (i < n1 && j < n2) {
        if (a[i] <= b[j]) {
            merged[k] = a[i];
            i++;
        } else {
            merged[k] = b[j];
            j++;
        }
    }
}
```

```

    }
    k++;
}

while (i < n1) {
    merged[k] = a[i];
    i++;
    k++;
}

while (j < n2) {
    merged[k] = b[j];
    j++;
    k++;
}

cout << "Merged sorted array: ";
for (int x = 0; x < k; x++) {
    cout << merged[x] << " ";
}
cout << endl;

return 0;
}

```

The screenshot shows a Windows terminal window with the following content:

```

PS C:\Users\jainm\OneDrive\Desktop\Striver_DSA> cd "c:\Users\jainm\OneDrive\Desktop\Striver_DSA\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter elements in sorted order:
1 2 3 4
Enter size of second sorted array: 3
Enter elements in sorted order:
4 5 6
Merged sorted array: 1 2 3 4 4 5 6
PS C:\Users\jainm\OneDrive\Desktop\Striver_DSA>

```

Q6. Write the above programs (1,2, and 3) using functions and call by address only.

```
#include <iostream>
using namespace std;

void linearSearch(int *arr, int n, int key) {
    int found = 0;
    for (int i = 0; i < n; i++) {
        if (*(arr + i) == key) {
            found = 1;
            cout << "Number found at position " << i + 1 << endl;
            break;
        }
    }
    if (found == 0) {
        cout << "Number not found in array" << endl;
    }
}

void secondMaxMin(int *arr, int n) {
    int max1 = -1000000, max2 = -1000000;
    int min1 = 1000000, min2 = 1000000;

    for (int i = 0; i < n; i++) {
        if (*(arr + i) > max1) {
            max2 = max1;
            max1 = *(arr + i);
        } else if (*(arr + i) > max2 && *(arr + i) != max1) {
            max2 = *(arr + i);
        }

        if (*(arr + i) < min1) {
            min2 = min1;
            min1 = *(arr + i);
        } else if (*(arr + i) < min2 && *(arr + i) != min1) {
            min2 = *(arr + i);
        }
    }

    cout << "Second maximum element is: " << max2 << endl;
    cout << "Second minimum element is: " << min2 << endl;
}
```

```
}
```

```
void insertElement(int *arr, int *n, int pos, int value) {  
    for (int i = *n; i >= pos; i--) {  
        *(arr + i) = *(arr + i - 1);  
    }  
    *(arr + pos - 1) = value;  
    *n = *n + 1;  
    cout << "Element inserted successfully" << endl;  
}
```

```
void deleteElement(int *arr, int *n, int pos) {  
    for (int i = pos - 1; i < *n - 1; i++) {  
        *(arr + i) = *(arr + i + 1);  
    }  
    *n = *n - 1;  
    cout << "Element deleted successfully" << endl;  
}
```

```
void traverseArray(int *arr, int n) {  
    cout << "Array elements: ";  
    for (int i = 0; i < n; i++) {  
        cout << *(arr + i) << " ";  
    }  
    cout << endl;  
}
```

```
int main() {  
    int arr[100];  
    int n;  
    cout << "Enter number of elements: ";  
    cin >> n;  
  
    for (int i = 0; i < n; i++) {  
        cout << "Enter element " << i + 1 << ": ";  
        cin >> arr[i];  
    }
```

```
    int key;  
    cout << "Enter number to search: ";  
    cin >> key;  
    linearSearch(arr, n, key);  
  
    secondMaxMin(arr, n);
```

```

int choice;
do {
    cout << "\n1. Insert\n2. Delete\n3. Traverse\n4. Exit\n";
    cout << "Enter your choice: ";
    cin >> choice;

    if (choice == 1) {
        int pos, value;
        cout << "Enter position to insert (1 to " << n + 1 << "): ";
        cin >> pos;
        cout << "Enter value to insert: ";
        cin >> value;
        insertElement(arr, &n, pos, value);
    }
    else if (choice == 2) {
        int pos;
        cout << "Enter position to delete (1 to " << n << "): ";
        cin >> pos;
        deleteElement(arr, &n, pos);
    }
    else if (choice == 3) {
        traverseArray(arr, n);
    }
    else if (choice == 4) {
        cout << "Exiting program" << endl;
    }
    else {
        cout << "Invalid choice" << endl;
    }

} while (choice != 4);

return 0;
}

```



```

Terminal
PS C:\Users\jainm\OneDrive\Desktop\Striver_DSA> cd "c:\Users\jainm\OneDrive\Desktop\Striver_DSA\" ; if ($?) { g++ tempCodeRunnerFile.cpp
p -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter number of elements: 4
Enter element 1: 2
Enter element 2: 4
Enter element 3: 7
Enter element 4: 8
Enter number to search: 5
Number not found in array

```